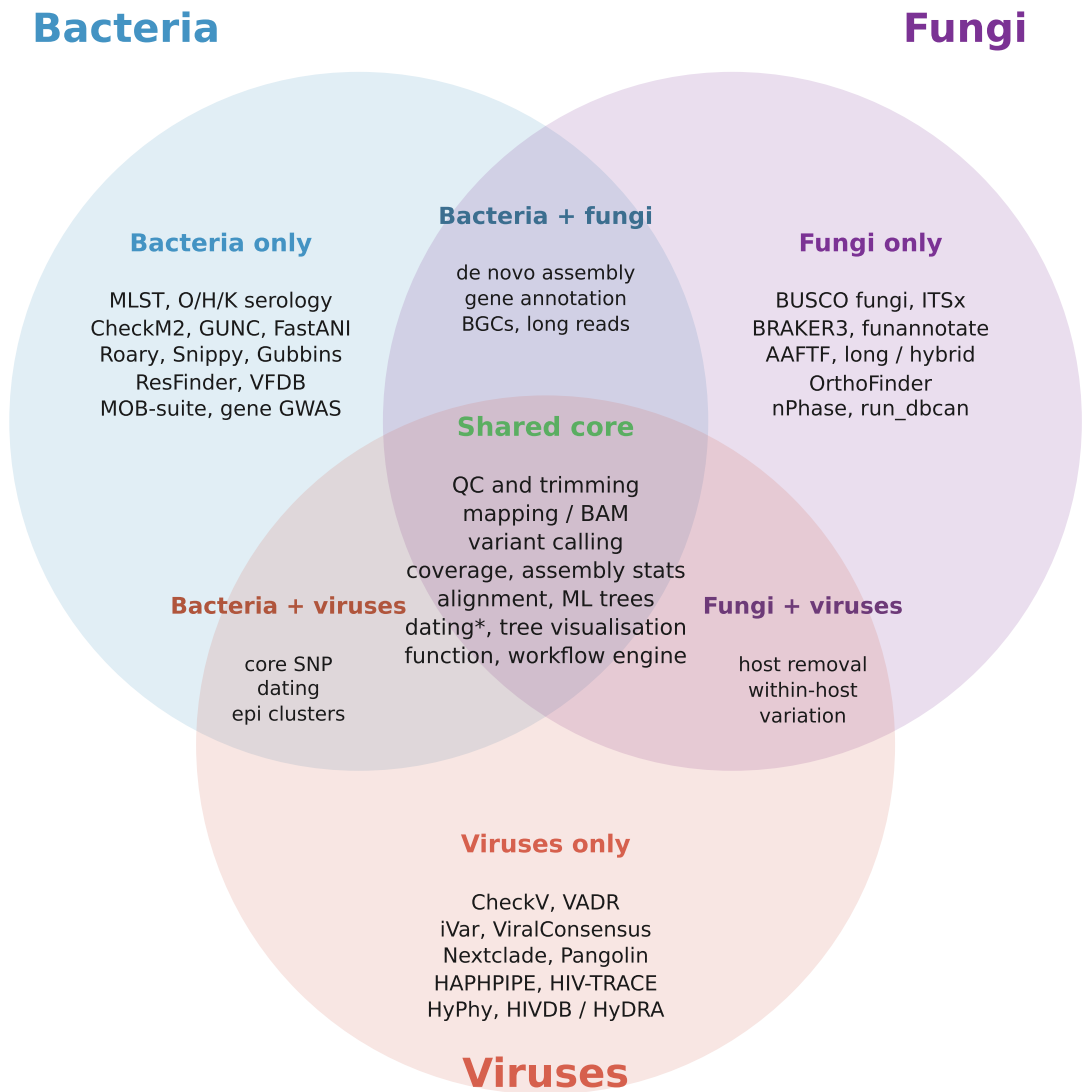


Extending EasyWGS beyond bacteria: shared core versus domain-specific tools

Bacterial isolate WGS, virus WGS (HIV, SARS-CoV-2, norovirus) and pathogenic fungal WGS (Aspergillus, Candida) share a domain-agnostic core but differ in assembly, completeness, annotation and typing.



* dating applies when a molecular clock / temporal signal exists (weak for many fungi).

Stage	Bacteria	Viruses	Fungi	Probiotic profile
Read QC	fastp, FastQC, MultiQC	fastp, FastQC, MultiQC	fastp, NanoPlot / chopper	same as bacteria
Host / contaminant removal	CLEAN, Kraken2, FCS-GX	map to host, KneadData, CheckV	map to host, BlobToolKit	same as bacteria
Assembly strategy	SPAdes, Unicycler, Flye	map + consensus (iVar, ViralConsensus); metaSPAdes for DNA	Flye / Canu / HybridSPAdes, NextPolish	same as bacteria
Completeness / QC gate	CheckM2, GUNC	CheckV	BUSCO fungi, FGMP	CheckM2
Gene annotation	Bakta / Prokka, Prodigal	VADR, VIGOR, VAPID	BRAKER3, funannotate, MAKER	Bakta + run_dbcan, antiSMASH
Species / strain identity	FastANI, mlst	VADR, lineage, Nextclade	ITSx / UNITE, TEF1, SNP, skani	FastANI, EFSA taxonomic unit
Typing / subtype	MLST, serotype, cgMLST	clade / subtype: Pangolin, Nextclade, HIV subtype, noro VP1/RdRp	SNP clade, ITS, MLST / MLVA	strain MLST
Resistance / virulence	abricate, RGI, ResFinder, VFDB	HIVDB / HyDRA, HyPhy selection	cyp51A / ERG11 / FKS SNPs, CNV / aneuploidy	acquired AMR absent and non-mobile, toxin absent (EFSA)
Pangenome / homology	Roary, Panaroo, PIRATE	whole-genome alignment, Nextalign (no gene pangenome)	OrthoFinder, GET_HOMOLOGUES	Panaroo
Variant / SNP calling	Snippy, Gubbins, bcftools	iVar / bcftools, low-frequency	GATK / freebayes, nPhase phasing	Snippy / bcftools
Phylogeny / dating	IQ-TREE 3, TreeTime, BactDating	Q-TREE, UShER, TreeTime, BEAST	IQ-TREE / RAXML-NG on orthologs	IQ-TREE, TreeTime
Genotype-phenotype GWAS	pyseer, Scoary, PLINK	limited: large N, strong linkage	hard: clonal, small samples	optional trait association

Probiotics are a use-profile, not a fourth domain: bacterial probiotics reuse the bacterial track with an EFSA QPS safety/benefit lens, while *Saccharomyces boulardii* follows the fungal track.